SE 423 Mechatronics

Laboratory Assignment #7

RCServo Test Program

TA Check-Offs in this document: 1 (marked "TA Check-Off" in the text and listed at the end)

Goals for this Lab Assignment

1. Learn where to connect RC Servos to the robot and drive them with similar code you developed in HW #4.

Exercise 1: Drive RC Servos connected to the Robot

Your instructor will take you through all the coding for this exercise. You will be using a new "Starter" project called "FinalProjectwithCANAprilStarter". You will be using this project starter in Lab 7 also.

The robot car can connect and drive five RC Servos (we will probably not use more than two). Your instructor will show you where to connect the RC Servos and that they are driven with PWM outputs EPWM3A, EPWM3B, EPWM5A, EPWM5B and EPWM6A. Perform the following steps to have the robot drive RC Servos:

1. In Timer2's ISR, (which is called by default every 40ms.) Just call the function:

```
PostSWI3();
```

   This will cause the `SWI3_LowestPriority` software interrupt to be called every 40ms. From here forward most of your code will need to be placed in the `SWI1_HighestPriority` function. You can also place less important code in the `SWI3_LowestPriority` function.

2. In `SWI3_LowestPriority`, first read the value of the encWheel:

```
RCangle = readEncWheel();
```

   Then call the five RC servo functions and pass `RCangle` to each of them:

```
setEPWM3A_RCServo(RCangle) //RCangle is a value from -90 to 90
setEPWM3B_RCServo(RCangle) //RCangle is a value from -90 to 90
setEPWM5A_RCServo(RCangle) //RCangle is a value from -90 to 90
setEPWM5B_RCServo(RCangle) //RCangle is a value from -90 to 90
setEPWM6A_RCServo(RCangle) //RCangle is a value from -90 to 90
```

3. Then print the value of `RCangle` to the robot's text LCD screen. Run your code. Now by spinning the encoder wheel in your hand, you can watch the text LCD screen and dial in the angle that moves the RC servos to certain angles.

   ☐ **TA Check-Off 1: All five RC servo connections**
   Show this code working to your Instructor and test that all 5 RC Servo connections work.

Check-Off Summary

Tick each box once your TA or instructor has checked you off.

- [ ] Check-Off 1: All five RC servo connections

Make sure that you comment you code and put your initials in your Gradescope submission! See https://marius-juston.github.io/SE-423---Class-Material/Homeworks/commenting_code_example.pdf for how we expect the code to be commented like!